Introduction to Cybersecurity and Ethical Hacking

Imagine this:

Sarah, a university student, just received an email that looked like it came from her bank. The message said there was a problem with her account and asked her to click a link and log in to fix it. Without thinking twice, Sarah clicked the link and entered her banking username and password.

A few days later, Sarah noticed several unauthorized transactions on her account. Someone had stolen her money. Shocked and frustrated, she contacted her bank. It turned out that the email was fake—a cyber attack designed to trick her into giving away sensitive information.

This is where **cybersecurity** comes in.



What is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, programs, and data from digital attacks, damage, or unauthorized access.

Why is Cybersecurity Important?

In today's world, almost everything is connected to the internet—banking, shopping, studying, healthcare, social media, and work. With so much valuable information online, cybercriminals are constantly looking for ways to exploit weaknesses.

Without proper cybersecurity:

- Personal data can be stolen.
- Financial loss can occur.
- Identities can be faked.
- Businesses can be shut down.
- National security can be threatened.

Why Businesses in Today's Era Need Cybersecurity

In today's digital world, businesses rely on technology more than ever. From storing customer data to making financial transactions, nearly everything is online. But as technology grows, so do the threats.

1. Protection of Sensitive Data

Businesses collect a lot of personal information:

- Customer names, emails, phone numbers
- Payment information (credit cards, bank details)
- Business secrets and financial records

If this data falls into the wrong hands, it can lead to:

- **Identity theft**
- **Financial fraud**
- **Loss of customer trust**

2. Rising Cybercrime Rates

Cyberattacks are increasing worldwide. Common threats include:

- **Phishing emails** that trick employees
- **Ransomware** that locks files until money is paid
- **Hacking** into weak systems to steal or damage data

3. Business Continuity

Cyberattacks can shut down business operations. Imagine:

- A company can't access its files.
- Customers can't make purchases online.
- Staff can't send emails or use systems.

This can cause **huge financial losses** and delay services.

4. Building Customer Trust

Customers want to know their data is safe. A business with strong cybersecurity shows it is:

- **Professional**
- **Trustworthy**
- **Responsible**

When people trust your security, they’re more likely to do business with you.

e.g MPESA

What is Ethical Hacking?

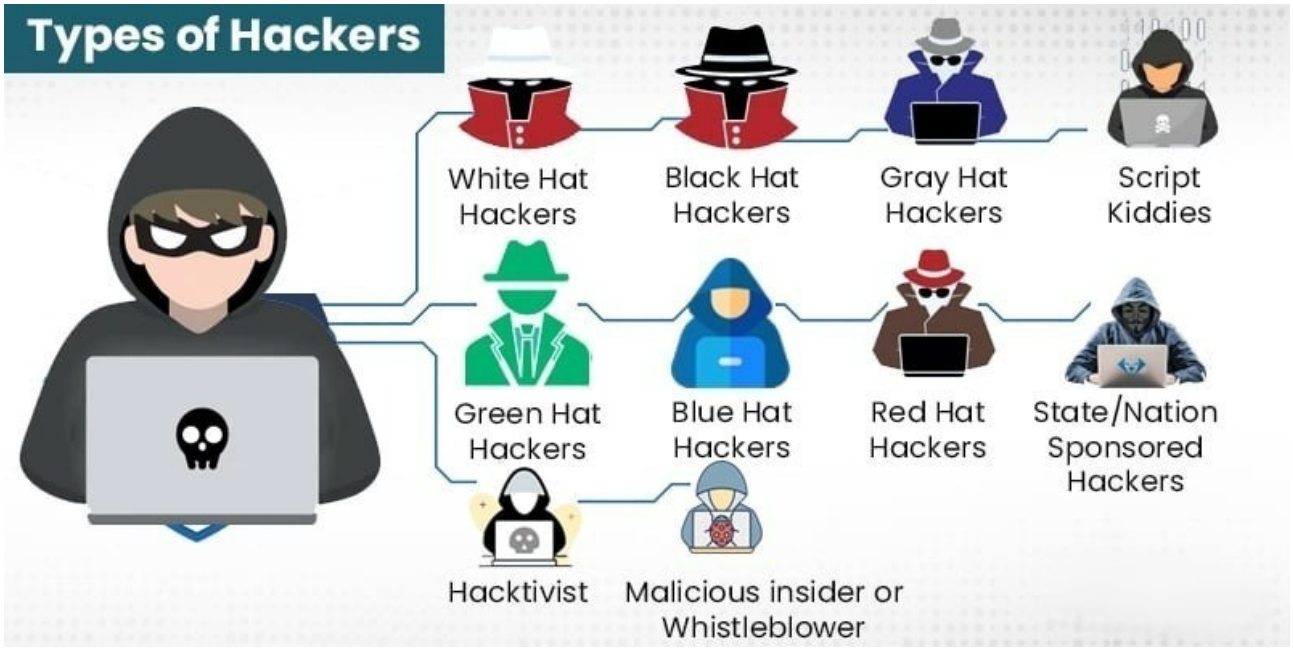
Definition

Ethical Hacking is the legal and authorized act of breaking into computers or systems to test their security and help improve/secure them.

Example

After Sarah's data was stolen, the bank took it seriously. They hired a cybersecurity expert —a hacker. But not the kind who steals information.

This hacker is known as an **ethical hacker**.



1. White Hat Hackers (Ethical Hackers)

These are **good hackers** who use their skills to protect systems and find security flaws before criminals can exploit them.

Example: A cybersecurity expert hired by a bank to test its system for weaknesses.

2. Black Hat Hackers (Criminal Hackers)

These are **malicious hackers** who break into systems to steal, damage, or make money illegally.

Example: A hacker who steals credit card information and sells it on the dark web.

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3. Grey Hat Hackers

They fall **between good and bad**. They may break into systems without permission but don't usually intend harm—and sometimes report the issues they find.

Example: A hacker who finds a bug in a company's website and reports it online or tells the website owner about the error.

5. State-Sponsored Hackers

These hackers work for a **government** to spy, disrupt, or attack other countries or organizations, often in secret.

Example: A cyber group that hacks into another country's defense system to gather intelligence.

Students research on others

Types of Cyber Attacks



Explanation for each type of cyberattack:

1. **SQL Injection Attack**
   A hacker enters **harmful commands** into a website to **mess with its database and steal data**

2. **Password Attack**
   Trying to guess or steal someone's password to access their accounts.

3. **Malware Attack**
   hacker **tricks you into installing bad software** (like a virus) that **steals or damages** your stuff.

4. **Birthday Attack**
   In hacking, this idea is used to find **two files with the same digital identity/name**
   If that happens, a hacker can **trick the computer into thinking a fake file is real**.

5. **Phishing Attack**
   A **fake email or message** that tries to **trick you into giving personal info**, like your password or credit card

6. **Drive-By Attack**
   You visit a website and **it secretly installs malware** on your device — **no clicking needed**.

7. **XSS (Cross-Site Scripting)**
   A hacker **adds bad code to a website** so when you visit, **your browser runs it**, possibly stealing your info.

8. **Man-in-the-Middle Attack**
   A hacker secretly intercepts and possibly alters the communication between two people or systems.

9. **DoS Attack (Denial of Service)**
   A hacker **floods a website with too much traffic**, so that the website slows and no one can access it,

10. Eavesdropping

It's when a hacker secretly listens to private conversations or data being shared between two people or systems. A hacker **just listens** to your conversation or data exchange.

Let’s Look at Some of These Attacks

While there are many different types of Cyberattacks, we’ll focus on a few of the most common ones.

1. Malware Attack?

A **malware attack** happens when harmful software, known as **malware – malicious software**, is secretly installed on your device. This software can **damage your system** or **steal your personal information**.

Types of Malware (Simple Overview):

- **Virus**: Like a **bug** that spreads from one file to another, damaging them along the way.
- **Spyware**: **Secretly watches** what you’re doing, stealing info like passwords, I.e **Keylogger**
- **Ransomware**: **Locks your files** and asks for money to unlock them.
- **Trojan Horse**: **Pretends to be helpful** software but actually causes harm once opened.

How Malware Gets Into Your Device:

- **Clicking on suspicious links** in emails or websites.
- **Downloading fake apps or files** from untrustworthy sources.
- **Infected USB drives** or devices.

How to Protect Yourself:

- **Install antivirus software.**
- **Be cautious with downloads and links.**
- **Update your software regularly.**

2. Phishing Attack

Phishing is when someone **pretends to be a trusted person or company** (like a bank or service) to **trick you into giving personal information**, such as passwords, PINs, or credit card numbers.

They often send **fake emails or messages** that look real — but are designed to **steal from you**.

Student Challenge: Spot the Phishing Red Flags

Here's a fake email filled with red flags. **Read it carefully** and see how many warning signs you can find!

Subject: Your WhatsApp Account Will Be Deactivated

From: support@whatsap-security.com

Hi,

We've detected **suspicious activity** on your WhatsApp account. To protect your account, we are planning to **deactivate it within 24 hours**.

To keep your account active, please:

Click here to verify your WhatsApp account

Failure to verify will result in **permanent account loss**, including all your chats and media.

For quicker support, reply with your **phone number and 6-digit verification code**.

Thank you,
WhatsApp Security Team

This is an automated message. Please do not ignore.

Your Task:

- What things look **suspicious or fake** in this email?
- Would you **trust this message** if it landed in your inbox?
- Make a list of at least **4 red flags** and explain why they're risky.

Solutions

Red Flags in the WhatsApp Phishing Email:

1. **Fake Sender Email Address**
    - **support@whatsap-security.com** — Notice the typo: **“Whatsap”** instead of **“WhatsApp”**.
    - **Why it’s suspicious**: Legitimate companies always use their official, correctly spelled domain name.

2. **Urgency and Threats**
    - The message states **“deactivate it within 24 hours”** or you’ll lose everything.
    - **Why it’s suspicious**: Phishing emails often create a **false sense of urgency** to pressure you into acting quickly without thinking.

3. **Suspicious Link**
    - **http://whatapp.net** — This is a **fake website**. Real WhatsApp links should end with **whatsapp.com**, not some random domain.
    - **Why it’s suspicious**: Legitimate websites always have official domains. A fake website is a common way to steal your data.

4. **Request for Personal Information**
    - The email asks you to **reply with your phone number and 6-digit verification code**.
    - **Why it’s suspicious**: Real companies never ask for sensitive information like this via email. This is a trick to steal your details.

5. **Generic Greeting**
    - The email simply starts with **“Hi”** instead of your actual name.
    - **Why it’s suspicious**: Real companies usually personalize emails by using your real name. Phishers often use vague greetings.

6. **Automated Language**
    - The line **“*This is an automated message. Please do not ignore*”** feels robotic and urgent.

- **Why it’s suspicious**: Phishing emails often use automated language to create pressure and trick you into clicking quickly.

Let’s Dive Into Kali Linux

Now that we’ve looked at some common cyberattacks, **we’ll start exploring Kali Linux** — a special operating system **used for hacking**.

But not just any hacking — Kali is mainly used by **ethical hackers**, who hack **legally** to test systems, find weaknesses, and help protect them from real attackers.

Next Lesson, we cover Ethical Hacking Practically on Kali Linux!

Take Away Assignment

Read on what is Kali Linux.

www.kali.org